Architecture sweep results

Signer-honest American Sign Language recognition · MacBook Pro, Apple Silicon MPS

Source: `experiments.ipynb` · *Split protocol C (cross-corpus) · 4 epochs · ≤400 images/class*

Status

The full zoo has **18** candidates. This report includes every result written to disk as of generation: **8 scored**, **1 in flight** (checkpoint present, no `result.json`), and **9 not started**. The leaderboard is marked `complete: false`. Selection is on validation macro-F1 only; the test split is reserved for the eventual winner and has not been used to rank architectures.

Experiment setup

Training images come from *asl_alphabet* (87,000 frames). The corpus is subsampled to at most 400 images per class by dropping whole recovered capture groups, never individual frames. Protocol C trains on that subsample and holds out *asl_alphabet_test* (a different person, room, and camera) as the test set. Validation is carved from the training corpus so the test corpus stays untouched until architecture selection is finished.

Split	Images	Groups	Classes	Role
train	10,795	1	29 / 29	fine-tune
val	805	105	29 / 29	select architecture / epoch
test	1,740	796	29 / 29	winner only, once

Device: MPS. Sample: 87,000 → 11,600 images. Hardware: Apple M4 Pro, 24 GB unified memory. Backbone search uses the same split for every candidate.

Protocol A / B / C (same backbone: vit_small)

Before the architecture search, the notebook ran three splits on `vit_small` (21.7M parameters, 4 epochs). The drop from A to C is the leakage story the rest of the sweep is built on.

Protocol	Train	Test	Test groups	Leak rate	Exact dups	Val F1	Test acc	Test F1
A random	8120	1740	40	0.9914	643	0.9994	1.0000	1.0000
B group	10795	402	52	0.0050	0	0.9952	0.6318	0.8163
C cross-corpus	10795	1740	796	0.0000	0	0.8939	0.6057	0.5684

Protocol A reports 100% test accuracy with a 99.14% leak rate and 643 exact duplicates in the test split. Protocol C has zero measured leak and 56.8% test macro-F1 on the same backbone — that is the number the architecture search is trying to beat, not the 100% from A.

Architecture leaderboard (validation only)

Candidates are ranked by the best validation macro-F1 across the four epochs. The highlighted row is the current leader among scored models (**deit3_base**). Yellow rows did not learn (validation accuracy near chance for 29 classes ≈ 3.4%). They are still on the board because the sweep records every finished run.

Rank	Preset	Family	Tier	Params (M)	px	Best ep	Val F1	Val acc	s / epoch
1	deit3_base	vit	arch	85.84	224	3	0.9456	0.8957	394.3
2	vit_base	vit	scale	85.82	224	1	0.9128	0.8870	381.3
3	beitv2_base	vit	arch	85.78	224	2	0.9080	0.8907	414.4
4	vit_small	vit	scale	21.68	224	2	0.8939	0.8149	121.6

Rank	Preset	Family	Tier	Params (M)	px	Best ep	Val F1	Val acc	s / epoch
5	clip_base	vit	arch	85.82	224	0	0.8886	0.8882	378.2
6	effnetv2_s	cnn	mobile	20.21	224	0	0.8426	0.7764	172.1
7	vit_tiny	vit	scale	5.53	224	3	0.0670	0.1021	2.8
8	efficientformer_s2	hybrid	mobile	12.15	224	3	0.0521	0.0957	378.4

Queue: remaining candidates

Preset	Family	Tier	≈ params (M)	img	Status	Note
swin_base	hybrid	arch	86.8	224	in flight — checkpoint only	hierarchical windowed attention
convnext_base	cnn	arch	87.6	224	not started	CNN control
convnextv2_base	cnn	arch	87.7	224	not started	CNN + MAE pretraining
siglip_base	vit	arch	92.9	224	not started	language-supervised (web images)
caformer_b36	hybrid	arch	95.8	224	not started	MetaFormer (conv then attention)
vit_large	vit	scale	303.3	224	not started	capacity ceiling; A100-class
mobilevit_v2	hybrid	mobile	9.8	256	not started	transformer blocks at mobile cost
eva02_base	vit	arch	86.4	448	not started	SOTA ceiling check at 448px
dinov2_small	vit	arch	22.1	518	not started	self-supervised, small scale
dinov2_base	vit	arch	86.6	518	not started	DINOv2 with registers

`swin_base` *has a* `best.pt` *under* `runs/C_arch_swin_base/` *but no result file yet, which matches a training run that has not finished writing its history.*

Per-model training logs

Each table is the four-epoch history from `runs/C_arch_<preset>/result.json`. Best is the epoch with the highest validation macro-F1 (that is also the checkpoint kept as `best.pt`). Test metrics are omitted on purpose: the sweep trains with `eval_test=False`.

vit_tiny · 5.53M params · vit/scale · 224px · best epoch 3 (val F1 0.0670, val acc 0.1021)

honest-accuracy floor

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	3.5829	4.27e-04	4.9	4.2042	0.0416	0.0270	
1	2.2104	2.50e-04	2.7	4.1079	0.0950	0.0642	
2	1.5830	7.32e-05	1.8	4.1240	0.0902	0.0572	
3	1.3693	0.00e+00	1.8	4.0671	0.1021	0.0670	yes

Wall time for 4 epochs: 11.2s (0.2 min). Mean 2.8s / epoch.

efficientformer_s2 · 12.15M params · hybrid/mobile · 224px · best epoch 3 (val F1 0.0521, val acc 0.0957)

on-device candidate

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	3.3142	4.27e-04	300.6	3.3674	0.0671	0.0305	
1	3.0742	2.50e-04	481.6	3.3562	0.0823	0.0335	
2	2.8186	7.32e-05	360.5	3.3402	0.0894	0.0378	
3	2.6933	0.00e+00	370.9	3.3222	0.0957	0.0521	yes

Wall time for 4 epochs: 1513.6s (25.2 min). Mean 378.4s / epoch.

effnetv2_s · 20.21M params · cnn/mobile · 224px · best epoch 0 (val F1 0.8426, val acc 0.7764)

CNN mobile control

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	1.6849	4.66e-04	175.2	0.9068	0.7764	0.8426	yes
1	0.7022	2.93e-04	172.5	0.6807	0.7963	0.8193	
2	0.6656	8.92e-05	172.1	0.8283	0.7950	0.7915	
3	0.6557	0.00e+00	168.4	0.7745	0.7950	0.7754	

Wall time for 4 epochs: 688.2s (11.5 min). Mean 172.1s / epoch.

vit_small · 21.68M params · vit/scale · 224px · best epoch 2 (val F1 0.8939, val acc 0.8149)

accuracy per FLOP in the ViT family

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	1.4832	2.80e-04	121.6	1.0885	0.8000	0.7820	
1	0.7298	1.76e-04	119.6	1.1863	0.7826	0.8063	
2	0.6814	5.34e-05	126.1	0.9327	0.8149	0.8939	yes
3	0.6647	0.00e+00	119.1	0.9439	0.8037	0.8278	

Wall time for 4 epochs: 486.4s (8.1 min). Mean 121.6s / epoch.

vit_base · 85.82M params · vit/scale · 224px · best epoch 1 (val F1 0.9128, val acc 0.8870)

standard research comparison point

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	1.5184	2.22e-06	378.4	0.8977	0.7677	0.8015	
1	0.7022	1.39e-06	387.7	0.6819	0.8870	0.9128	yes
2	0.6649	4.24e-07	378.8	0.8110	0.8075	0.8455	
3	0.6596	0.00e+00	380.3	0.7663	0.8261	0.8574	

Wall time for 4 epochs: 1525.2s (25.4 min). Mean 381.3s / epoch.

deit3_base · 85.84M params · vit/arch · 224px · best epoch 3 (val F1 0.9456, val acc 0.8957)

same architecture as ViT-B, better recipe

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	1.5390	2.22e-06	375.5	0.6619	0.8522	0.8738	
1	0.7260	1.39e-06	394.1	0.5213	0.8894	0.9355	
2	0.6846	4.24e-07	404.0	0.5239	0.8957	0.9397	
3	0.6734	0.00e+00	403.5	0.4998	0.8957	0.9456	yes

Wall time for 4 epochs: 1577.1s (26.3 min). Mean 394.3s / epoch.

beitv2_base · 85.78M params · vit/arch · 224px · best epoch 2 (val F1 0.9080, val acc 0.8907)

masked image modelling

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	1.7141	2.22e-06	410.6	0.5639	0.8571	0.8614	
1	0.6681	1.39e-06	420.6	0.5391	0.8919	0.9010	
2	0.6520	4.24e-07	421.7	0.5723	0.8907	0.9080	yes
3	0.6521	0.00e+00	404.7	0.5299	0.8907	0.8978	

Wall time for 4 epochs: 1657.6s (27.6 min). Mean 414.4s / epoch.

clip_base · 85.82M params · vit/arch · 224px · best epoch 0 (val F1 0.8886, val acc 0.8882)

LAION-2B language-supervised

Epoch	Train loss	LR	Seconds	Val loss	Val acc	Val F1	Best?
0	1.4002	2.22e-06	378.3	0.5743	0.8882	0.8886	yes
1	0.6873	1.39e-06	394.3	0.7974	0.8012	0.8568	
2	0.6604	4.24e-07	385.4	0.8826	0.7988	0.8344	
3	0.6550	0.00e+00	354.9	0.8631	0.8000	0.8417	

Wall time for 4 epochs: 1512.9s (25.2 min). Mean 378.2s / epoch.

Reading notes

Current leader among scored models: DeiT III base, 94.56% validation macro-F1 and 89.57% validation accuracy at epoch 3. It is the only scored backbone whose validation F1 still rose on the last epoch, so four epochs may be short for it.

Scale ladder so far: vit_tiny failed to train (10.2% val acc, 2.8s/epoch — not a real training run at this data size). vit_small 0.894 F1, vit_base 0.913 F1. Capacity helps on this validation split, but the honest number is still the eventual cross-corpus test, which has not been run for anyone except the earlier vit_small protocol-C baseline (test F1 0.568).

Failed runs: vit_tiny and efficientformer_s2 sit near chance. efficientformer_s2 did spend ~25 minutes per epoch, so it is not a timing glitch; the loss barely moved. Do not treat those two rows as architecture findings.

Overfitting on CLIP: clip_base peaked at epoch 0 (val F1 0.889) and then lost about 5 F1 points. effnetv2_s did the same (best at epoch 0). Early-stopped checkpoints are what the leaderboard uses.

What this PDF is not: a final architecture decision. Ten candidates are still outstanding, including the CNN controls (ConvNeXt / ConvNeXt V2), the self-supervised entries (DINOv2, EVA-02), SigLIP, CAFormer, MobileViT v2, ViT-Large, and Swin (in flight). Re-generate this file after the sweep writes `complete: true`.